Fiscal Year:	FY 2004	**Task Last Updated:**	FY 08/31/2004
PI Name:	Smith, Scott M Ph.D.		
Project Title:	Calcium Kinetics During Spaceflight		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	scott.m.smith@nasa.gov	**Fax:**	FY 281-483-2888
PI Organization Type:	NASA CENTER	**Phone:**	281-483-7204
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Biomedical Research and Environmental Sciences Division/SK3		
PI Address 2:	2101 NASA Pkwy		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	06/01/1998	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	McCollum, Suzanne	**Contact Phone:**	281 483-7307
Contact Email:	suzanne.g.mccollum@nasa.gov		
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-107, R2		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	O'Brien, Kimberly (Johns Hopkins University School of Public Health) Abrams, Steven (Baylor College of Medicine) Wastney, Meryl (Metabolic Modeling Services)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Bone loss is one of the most detrimental effects of space flight, threatening to limit the duration of human space missions. The ability to understand and counteract this loss will be critical for crew health and safety during and after extended-duration missions. The hypotheses to be tested in this project are (1) that space flight alters calcium homeostasis and bone mineral metabolism, and (2) that calcium homeostasis and bone mineral metabolism will return to baseline within days to weeks of return to Earth. The first hypothesis will be evidenced by elevated rates of bone mineral resorption and decreased bone mineral deposition, decreased absorption of dietary calcium, altered calcitropic endocrine profiles, elevated excretion of calcium in urine and feces, and elevated excretion of markers of bone resorption. The second hypothesis will be evidenced by return of indices of calcium homeostasis and bone metabolism to preflight levels within days to weeks of return to Earth. Studies will be conducted on Shuttle and International Space Station astronauts before, during, and after short-duration and extended-duration flights. Measurements of calcium kinetics, bone mass, and endocrine/biochemical markers of bone and calcium homeostasis will be conducted. Kinetic studies utilizing isotopic tracers and mathematical modeling techniques will allow for determination of bone calcium deposition, bone calcium resorption, dietary calcium absorption and calcium excretion (both urinary and endogenous fecal excretion). These studies will build upon preliminary work conducted on the Russian Mir space station. The results from this project will be critical for determining the nature of effects of microgravity on bone and calcium homeostasis, as well as in determining the time course of these changes. Furthermore, this will provide an important baseline measurement for future assessments of countermeasure efficacy. These results are expected to aid in developing countermeasures for bone loss, both for space crews and for individuals on Earth who have metabolic bone diseases.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	While this study will obtain valuable information that will ensure crew health and safety, it will be important in understanding bone metabolism and in developing treatment for bone diseases in the general population. Osteoporosis is a skeletal disease characterized by bone loss and increased fracture risk. An estimated 28 million Americans either have osteoporosis or are at high risk of developing the disease. A better understanding of the impact of weightlessness on bone metabolism will allow for a better understanding of bone physiology. Furthermore, the techniques involved in this study are also employed in research of difficult to study populations (e.g., during pregnancy, infants, children, elderly). Modifications to these techniques for use inflight have already been employed in general research projects by other scientists (e.g., the use of saliva samples to minimize blood). Other techniques for sample preservation during space flight have been developed and patented. This will have applications in rural medicine, where samples could be collected at home and mailed to the laboratory for analysis. Additionally, beyond the more direct benefits of the science, this Experiment Team has worked extensively to use this as a valuable educational/outreach opportunity, to interest students of all ages in science, and also to provide general health information, with the backdrop of spaceflight to obtain and maintain interest.
Task Progress:	Following the release of the CAIB report, embargoed preflight samples were available for analysis. We are currently completing these analyses and hope to begin statistical analysis in mid-2004.
Bibliography Type:	Description: (Last Updated: 01/19/2026)
Articles in Peer-reviewed Journals	Smith SM, Dillon EL, DeKerlegand DE, Davis-Street JE. "Variability of collagen crosslinks: Impact of sample collection period." Calcif Tissue Int. 2004 Apr;74(4):336-41. PMID: 15255070 , Apr-2004
Articles in Peer-reviewed Journals	Smith MD, Davis-Street JE, Calkins DS, Nillen JL, Smith SM. "Stability of i-STAT EC6+ cartridges: effect of storage temperature on shelf life." Clin Chem. 2004 Mar;50(3):669-73. PMID: 14981042 , Mar-2004
Articles in Peer-reviewed Journals	Van Heel N, Pettit J, Rice BL, Smith SM. "Meeting expanding needs for nutrient specificity: The NASA case study." Journal of Food Composition and Analysis 2003 Jun;16(3):299-303. http://dx.doi.org/10.1016/S0889-1575(03)00047-4 , Jun-2003
Articles in Peer-reviewed Journals	Smith SM, Wastney ME, O'Brien KO, Morukov BV, Larina IM, Abrams SA, Davis-Street JE, Oganov V, Shackelford LC. "Bone markers, calcium metabolism, and calcium kinetics during extended-duration space flight on the mir space station." J Bone Miner Res. 2005 Feb;20(2):208-18. Epub 2004 Nov 8. PMID: 15647814 , Feb-2005